

 Text Functions & Extraction XLOOKUP & INDEX/MATCH

INSTANT

 Data Analysis Diploma • Session 05

# ADVANCED FUNCTIONS

Text Functions · Lookup Functions · Dynamic Arrays · Power Query  
Fundamentals

**6 Goals****Core Outcomes**

Text • Lookups • Power Query

**44 Slides****Curriculum Depth**

Formulas &amp; Power Query

**Visual Labs****Hands-on Practice**

Lookup Simulator &amp; PQ

Start Session 05 →

 Dynamic Array Formulas Power Query ETL Pipelines

— SESSION COMPETENCIES

# What You Will Master Today

By the end of this session, you will confidently apply these advanced Excel and ETL capabilities:



01

## Master Text Extraction

Extract, clean, and reformat strings using LEFT, RIGHT, MID, LEN, FIND, and SUBSTITUTE.



02

## Advanced Lookups

Connect datasets seamlessly with XLOOKUP, VLOOKUP, HLOOKUP, and INDEX/MATCH.



03

## Dynamic Array Formulas

Build self-updating reports using FILTER, UNIQUE, SORT, and SORTBY functions.



04

## Power Query Fundamentals

Automate repetitive data cleaning tasks without writing a single line of code.



05

## ETL Data Pipeline

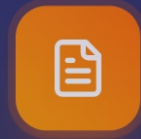
Import, clean, transform, and load messy data into analytical models with 1-click refresh.



06

## Real Business Scenarios

Construct automated sales reporting dashboards, HR directories, and inventory logs.



PART 1

# Excel Text Functions

Extracting, Parsing, Standardizing, and Combining Text Data like a  
Professional Analyst

— STRING PARSING

# LEFT & RIGHT Functions

Extract characters from the beginning or end of a text string.



## KEY CONCEPTS & SYNTAX

- LEFT extracts a specified number of characters from the start of text.
- RIGHT extracts characters from the end of text.
- Essential for splitting product codes, IDs, zip codes, or names.
- Syntax: =LEFT(text, num\_chars) | =RIGHT(text, num\_chars)
- Common Mistake: Specifying more characters than exist in the source cell.

## EXCEL FORMULA SYNTAX

```
=LEFT(A2, 3)
```

```
=RIGHT(A2, 4)
```

## REAL BUSINESS EXAMPLE

Extract product category code "ELEC" and product serial "2045" from SKU "ELEC-2045".

— MIDDLE EXTRACTION & LENGTH

# MID & LEN Functions

Extract substring from the middle of text and measure string length.



## KEY CONCEPTS & SYNTAX

- MID extracts characters from inside a text string starting at a specific position.
- LEN returns the total character count of a text string.
- MID requires: text, start\_num, and num\_chars.
- LEN is ideal for data validation (e.g. verifying 10-digit phone numbers or 14-digit IDs).
- Common Mistake: Incorrect start position skips or grabs wrong characters.

## EXCEL FORMULA SYNTAX

```
=MID(A2, 6, 4)
```

```
=LEN(A2)
```

## REAL BUSINESS EXAMPLE

Extract the year "2045" from SKU "ELEC-2045-A" and validate ID length.

— TEXT CONCATENATION

# CONCAT & TEXTJOIN Functions

Combine multiple text strings into a single formatted value.



## KEY CONCEPTS & SYNTAX

- CONCAT joins text from multiple cells or ranges without delimiters.
- TEXTJOIN joins text with a chosen delimiter (comma, space, dash).
- TEXTJOIN automatically ignores empty cells when ignore\_empty is set to TRUE.
- Great for assembling full names, complete mailing addresses, or combined tags.
- Common Mistake: Forgetting delimiter argument or quotes in TEXTJOIN.

## EXCEL FORMULA SYNTAX

```
=CONCAT(B2, C2)
```

```
=TEXTJOIN(", ", TRUE, D2:D5)
```

## REAL BUSINESS EXAMPLE

Combine First Name and Last Name columns into one Full Name, or concatenate address lines.

— TEXT FORMATTING

# UPPER, LOWER & PROPER

Standardize letter casing across inconsistent customer and product records.



## KEY CONCEPTS & SYNTAX

- UPPER converts all characters in a text string to UPPERCASE.
- LOWER converts all characters to lowercase.
- PROPER capitalizes the first letter of each word and lowers all other letters.
- Standardizes inconsistent manual data entry from web forms or CRM exports.
- Common Mistake: Applying to numeric values or expecting automatic live cell updates.

## EXCEL FORMULA SYNTAX

`=UPPER(E2)`

`=LOWER(E2)`

`=PROPER(E2)`

## REAL BUSINESS EXAMPLE

Standardize customer names entered inconsistently ("jOHn sMItH" -> "John Smith").

— WHITESPACE REMOVAL

# TRIM Function

Remove excess spaces from text while keeping single spaces between words.



## KEY CONCEPTS & SYNTAX

- Removes all leading spaces, trailing spaces, and multiple spaces between words.
- Keeps a single space character between words for proper formatting.
- Essential for cleaning imported CSV, web data, or copied database extracts.
- Fixes mysterious lookup failures caused by invisible trailing spaces.
- Common Mistake: Assuming TRIM removes non-breaking spaces (CHAR160) without SUBSTITUTE.

## EXCEL FORMULA SYNTAX

```
=TRIM(F2)
```

## REAL BUSINESS EXAMPLE

Clean up messy email addresses copied from webforms ("john.doe@email.com " -> "john.doe@email.com").

— POSITION SEARCH

# FIND Function

Locate the exact character position of a substring within text.



## KEY CONCEPTS & SYNTAX

- Locates the starting position of a target substring within a text string.
- Case-sensitive search function (unlike SEARCH which is case-insensitive).
- Returns a numeric position integer (e.g. 5 means the character is at position 5).
- Frequently combined with MID or LEFT for dynamic, variable-length text extraction.
- Common Mistake: Text not found returns a #VALUE! error (wrap with IFERROR or ISNUMBER).

## EXCEL FORMULA SYNTAX

```
=FIND("-", G2)
```

## REAL BUSINESS EXAMPLE

Locate the dash position within an order reference number ("ORD-99214") to dynamically split IDs.

— STRING REPLACEMENT

# SUBSTITUTE Function

Replace specific text within a string programmatically.



## KEY CONCEPTS & SYNTAX

- Replaces existing text with new text in a text string.
- Can target all occurrences or a specific occurrence instance if needed.
- Ideal for correcting formatting errors, changing date separators, or stripping unwanted characters.
- Different from Find & Replace tool because it updates dynamically via formula.
- Common Mistake: Case sensitivity causes missed replacements if case doesn't match.

## EXCEL FORMULA SYNTAX

```
=SUBSTITUTE(H2, "/", "-")
```

```
=SUBSTITUTE(H2, " ", "", 1)
```

## REAL BUSINESS EXAMPLE

Standardize date separators across imported reports  
("12/05/2024" -> "12-05-2024").

— NESTED FORMULAS

# Combining MID + FIND + LEN

Extracting variable-length sub-strings dynamically when positions change.

## 1. FIND DELIMITER

FIND("-", A2) locates the hyphen position in string "PROD-88219-US".

`FIND("-", A2) -> 5`

## 2. MID EXTRACTION

MID starts at position 6 (5 + 1) and extracts 5 digits.

`MID(A2, 6, 5) -> "88219"`

## 3. WRAP WITH TRIM

TRIM ensures no hidden spaces remain after extraction.

`TRIM(MID(A2, FIND("-", A2)+1, 5))`

**PRACTICAL APPLICATION:** Extract product serial numbers of varying lengths from non-standardized invoice codes.

— LAB 01

# Interactive Text Cleaning Studio

Test text functions live on messy raw datasets and observe output transformations.

## 1. SELECT INPUT SAMPLE

Messy SKU

" ELEC-2045-US "

Unformatted Name

"jOHn.sMItH@company.COM "

Invoice Code

"ORD/2024/9912"

## 2. SELECT FUNCTION

TRIM

UPPER

LOWER

PROPER

LEFT (4)

RIGHT (4)

SUBSTITUTE (/) to (-)

## 3. LIVE EXCEL TRANSFORMATION RESULT

RAW CELL VALUE (A2)

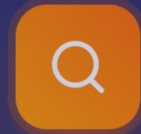
" ELEC-2045-US "

EVALUATED FORMULA

=TRIM(A2)

TRANSFORMED RESULT OUTPUT:

"ELEC-2045-US"



PART 2

# Lookup & Reference Functions

Searching, Connecting, and Retrieving Data Across Tables with VLOOKUP,  
HLOOKUP, XLOOKUP & INDEX/MATCH

— CLASSIC LOOKUP

# VLOOKUP Function

Search for a value in the first column of a table and return a value in the same row from another column.



## KEY CONCEPTS & SYNTAX

- Searches for a value in the FIRST column of a table range.
- Returns a matching value from a specified column index number.
- Requires setting range\_lookup to FALSE (or 0) for exact matches.
- Widely used across business models, but restricted to left-to-right lookups only.
- Common Mistake: Forgetting FALSE/0 defaults to TRUE (approximate match), returning wrong data!

## EXCEL FORMULA SYNTAX

```
=VLOOKUP(I2, ProductTable, 3, FALSE)
```

## REAL BUSINESS EXAMPLE

Look up product unit price using Product ID from a master pricing table.

— ROW-BASED LOOKUP

# HLOOKUP Function

Search for a value in the top row of a table and return a value in the same column from a specified row.



## KEY CONCEPTS & SYNTAX

- Searches for a value in the FIRST row of a horizontal data table.
- Returns a matching value from a specified row index number.
- Used when data headers are arranged vertically along rows.
- Follows exact same match logic as VLOOKUP (set FALSE for exact match).
- Common Mistake: Attempting to use HLOOKUP on standard vertical datasets.

## EXCEL FORMULA SYNTAX

```
=HLOOKUP(J2, SalesTable, 4, FALSE)
```

## REAL BUSINESS EXAMPLE

Find quarterly revenue from a horizontally structured finance summary sheet.

— RECOMMENDED STANDARD

# XLOOKUP Function

The modern, flexible replacement for VLOOKUP and HLOOKUP in Excel 365.



## KEY CONCEPTS & SYNTAX

- Modern replacement that supersedes VLOOKUP, HLOOKUP, and INDEX/MATCH.
- Searches left, right, up, or down — in any direction without column index numbers.
- Defaults to exact match automatically (no need to type FALSE).
- Includes built-in "if\_not\_found" error handling parameter to prevent #N/A errors.
- Common Mistake: Trying to use XLOOKUP in older legacy versions of Excel (pre-2021/365).

## EXCEL FORMULA SYNTAX

```
=XLOOKUP(K2, IDRange, NameRange, "Not Found")
```

## REAL BUSINESS EXAMPLE

Retrieve employee names using employee IDs, even when ID column is to the right of Name column.

— POWERFUL PAIR

# INDEX & MATCH Combination

The classic high-performance alternative for complex lookups in any version of Excel.



## KEY CONCEPTS & SYNTAX

- INDEX returns a value at a given row and column position in a range.
- MATCH finds the row position number of a search value in a column.
- Combined, MATCH finds the row number and INDEX pulls the corresponding data.
- Can look up in any direction (left or right) and does not break when columns are inserted.
- Common Mistake: Mismatched range sizes between INDEX return range and MATCH lookup range.

## EXCEL FORMULA SYNTAX

```
=INDEX(NameRange, MATCH(L2, IDRange, 0))
```

## REAL BUSINESS EXAMPLE

Retrieve a project manager's name based on Project ID when columns are reorganized.

METHOD EVALUATION

# Lookup Function Comparison

Choose the right lookup methodology based on data structure and Excel environment.

| Feature                 | VLOOKUP                      | HLOOKUP                      | XLOOKUP ★                          | INDEX & MATCH                  |
|-------------------------|------------------------------|------------------------------|------------------------------------|--------------------------------|
| Lookup Direction        | Left-to-Right only           | Top-to-Bottom only           | Any direction (Left/Right/Up/Down) | Any direction                  |
| Match Default           | Approximate (Requires FALSE) | Approximate (Requires FALSE) | Exact Match by Default             | Requires 0 for Exact Match     |
| Column Insertion Safety | Breaks (Static column index) | Breaks (Static row index)    | Safe (Dynamic array reference)     | Safe (Dynamic array reference) |
| Built-in Error Handling | No (Requires IFERROR)        | No (Requires IFERROR)        | Yes (if_not_found argument)        | No (Requires IFERROR)          |
| Excel Version           | All Excel Versions           | All Excel Versions           | Excel 365 / 2021+                  | All Excel Versions             |

**BUSINESS ADVICE:** A financial services firm migrates legacy VLOOKUP reports to XLOOKUP for speed and reliability.

## CLEAN REPORTS

# Shielding Lookups with IFERROR

Prevent ugly #N/A, #VALUE!, and #REF! errors from spoiling executive dashboards.



### KEY CONCEPTS & SYNTAX

- IFERROR catches any formula error and replaces it with a clean fallback value or text.
- IFNA specifically catches #N/A missing match errors while letting other calculation errors surface.
- Prevents downstream calculations (SUM, AVERAGE) from breaking due to a single missing match.
- Essential when looking up newly added products or customer IDs that may not exist yet.
- Common Mistake: Wrapping formulas with IFERROR too early without fixing broken formula syntax.

### EXCEL FORMULA SYNTAX

```
=IFERROR(VLOOKUP(A2, ProductTable, 3, FALSE), "Not Found")
```

```
=IFNA(XLOOKUP(A2, IDRange, PriceRange), 0)
```

### REAL BUSINESS EXAMPLE

Replace "#N/A" errors with "Item Discontinued" or 0 in sales invoice lookup tables.

— TWO-WAY MATCHING

# Two-Way Matrix Lookup (INDEX + 2 MATCHes)

Looking up values across both rows and columns simultaneously in a matrix.

## 1. ROW MATCH

MATCH(Region, RegionList, 0) finds the row number for "North".

```
MATCH("North", A2:A10, 0) -> Row 3
```

## 2. COLUMN MATCH

MATCH(Month, MonthList, 0) finds the column number for "March".

```
MATCH("March", B1:M1, 0) -> Col 3
```

## 3. INDEX MATRIX FETCH

INDEX returns value at intersection of Row 3 and Col 3.

```
INDEX(B2:M10, 3, 3)
```

**PRACTICAL APPLICATION:** Retrieve monthly budget figures from a 2D matrix matching both Region and Month.

SIMULATOR

# Interactive Lookup Algorithm Simulator

Compare execution of VLOOKUP vs XLOOKUP vs INDEX/MATCH on a live product database.

1. SELECT TARGET ID

P-101

P-102

P-103

P-104

2. SELECT LOOKUP METHOD

XLOOKUP

VLOOKUP

INDEX/MATCH

LIVE PRODUCT TABLE DATA

MATCHING ROW #2

| Row # | Product ID (Col A) | Product Name (Col B) | Price (Col C) |
|-------|--------------------|----------------------|---------------|
| 1     | P-101              | MacBook Pro 16"      | \$2,499       |
| 2     | P-102              | Dell XPS 15          | \$1,899       |
| 3     | P-103              | iPad Pro 12.9"       | \$1,099       |
| 4     | P-104              | Logitech MX Master   | \$99          |

EXECUTED EXCEL FORMULA:

=XLOOKUP("P-102", A2:A5, C2:C5) -> \$1,899



PART 3

# Dynamic Array Functions

Next-Generation Filtering, Sorting & Distinct Extraction with Modern Excel  
Dynamic Engines

— AUTO-SPILL

# FILTER Function

Extract rows that meet specified logical criteria dynamically.

## KEY CONCEPTS & SYNTAX

- Returns an array of rows that satisfy one or more logical criteria.
- Dynamically spills results into adjacent empty cells automatically.
- No need for manual filter menus, copy-pasting, or complex array formulas.
- Updates live whenever source data or filter criteria change.
- Common Mistake: Criteria range length not matching data range height causes #VALUE! error.

## EXCEL FORMULA SYNTAX

```
=FILTER(OrdersTable, Region="East")
```

```
=FILTER(OrdersTable, (Region="East")*(Sales>5000),  
"No Records")
```

## REAL BUSINESS EXAMPLE

Instantly display all high-value orders placed from the "East" region.

— DUPLICATE STRIPPING

# UNIQUE Function

Extract distinct values from a column or range automatically.



## KEY CONCEPTS & SYNTAX

- Extracts unique, distinct values from a list or multi-column range.
- Removes duplicate entries automatically without altering source data.
- Ideal for building clean data validation drop-down lists dynamically.
- Combines seamlessly with SORT for organized, alphabetical outputs.
- Common Mistake: Expecting UNIQUE to modify the original source dataset.

## EXCEL FORMULA SYNTAX

```
=UNIQUE(CustomerList)
```

```
=SORT(UNIQUE(CategoryColumn))
```

## REAL BUSINESS EXAMPLE

Generate a clean list of unique customer names from a raw orders log of 10,000+ rows.

— AUTO-RANKING

# SORT Function

Arrange data in ascending or descending order dynamically.



## KEY CONCEPTS & SYNTAX

- Sorts the contents of a range or array based on a specified column index.
- Supports ascending (1) and descending (-1) sort directions.
- Dynamic — automatically re-sorts output when underlying source numbers update.
- Does not alter or touch original data rows.
- Common Mistake: Sorting only one column of a multi-column table without selecting full range.

## EXCEL FORMULA SYNTAX

```
=SORT(SalesTable, 3, -1)
```

## REAL BUSINESS EXAMPLE

Automatically rank sales representatives by revenue from highest to lowest.

— FLEXIBLE SORTING

# SORTBY Function

Sort a range based on values in another column or external array.



## KEY CONCEPTS & SYNTAX

- Sorts a range based on another corresponding range or array values.
- Allows sorting by columns that are not included in the final output array.
- Supports multiple sort levels (e.g. Sort by Department ascending, then Salary descending).
- Provides greater flexibility than standard SORT for custom matrix reports.
- Common Mistake: Mismatched array lengths between target range and sort-by range.

## EXCEL FORMULA SYNTAX

```
=SORTBY(ProjectList, PriorityLevel, -1)
```

```
=SORTBY(EmpNames, DeptList, 1, SalaryList, -1)
```

## ||| REAL BUSINESS EXAMPLE

Sort project tasks by priority level score, rather than alphabetically by task name.

— POWER TRIAD

# Combining **FILTER + UNIQUE + SORT**

Constructing powerful single-cell dynamic reports that update in real time.

## 1. FILTER DATA

`FILTER(CustomerNames, Status="Active")` returns active customer names.

```
FILTER(A2:A100, B2:B100="Active")
```

## 2. UNIQUE EXTRACTION

`UNIQUE` strips away duplicate customer orders to get distinct names.

```
UNIQUE(FILTER(...))
```

## 3. SORT ALPHABETICALLY

`SORT` arranges the unique active customer list alphabetically.

```
=SORT(UNIQUE(FILTER(A2:A100,  
B2:B100="Active")))
```

**PRACTICAL APPLICATION:** Generate an automated, self-updating dropdown list of active unique client names.

CASE STUDY 01

# Dynamic Sales Reporting Dashboard

Combine XLOOKUP + FILTER + TEXTJOIN for live executive dashboard automation.

## IMPLEMENTATION STEPS

- ✓ Combine XLOOKUP and FILTER for an auto-updating live sales dashboard.
- ✓ Pull product unit prices automatically into raw orders upon transaction.
- ✓ Filter live results dynamically by Region and Sales Representative.
- ✓ Use TEXTJOIN to build combined summary order logs for management.
- ✓ Deliver a live, self-updating report that eliminates monthly re-work.

## BUSINESS IMPACT & OUTCOME

**"Sales dashboard auto-updates instantly as new order rows are appended."**

⚡ Zero manual copy-pasting required on recurring reports.

## — CASE STUDY 02

# HR Searchable Employee Directory

Clean names, look up departments, and extract distinct teams for HR management.

## IMPLEMENTATION STEPS

- ✓ Use CONCAT / TEXTJOIN to build standardized full employee names.
- ✓ Use PROPER to correct inconsistent name capitalization from job applications.
- ✓ Use INDEX / MATCH to pull department and manager names from org charts.
- ✓ Use UNIQUE to list all distinct departments for filter controls.
- ✓ Build a clean, searchable HR directory for 300+ staff members.

## BUSINESS IMPACT & OUTCOME

**"HR team saves 5+ hours per week managing employee onboarding logs."**

⚡ Zero manual copy-pasting required on recurring reports.

## — CASE STUDY 03

# Automated Warehouse Inventory Control

Filter low-stock items and rank by reorder urgency automatically.

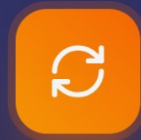
## IMPLEMENTATION STEPS

- ✓ Use FILTER to isolate low-stock items where Stock < Minimum Threshold.
- ✓ Use SORT to rank items by reorder urgency score (highest priority first).
- ✓ Use SUBSTITUTE to clean messy SKU formatting across legacy warehouses.
- ✓ Use UNIQUE to list distinct product categories needing immediate restocking.
- ✓ Support faster, error-free warehouse decision-making and automated PO generation.

## BUSINESS IMPACT & OUTCOME

**"Warehouse logistics team gets a live low-stock priority list updated every morning."**

⚡ Zero manual copy-pasting required on recurring reports.



PART 4

# Power Query Introduction

Connecting, Cleaning, and Reshaping Data Without Writing Code

— FOUNDATION

# What is Power Query?

A built-in Excel tool that connects, cleans, and reshapes data before it ever hits a worksheet.



## Import Data

Connect to files (CSV, Excel), folders, databases & web APIs.



## Clean Data

Remove errors, blank rows, extra spaces & inconsistent values.



## Transform Data

Reshape rows & columns into the exact layout needed for analysis.



## Load Data

Deliver clean, ready-to-use tables directly into Excel or Data Model.



### DID YOU KNOW?

Power Query automatically records every transformation step you perform — so you can review, edit, or reuse the exact same cleaning process next time with a single click of Refresh!

— WHY POWER QUERY

# Why Use Power Query?

The same clean-up work, two very different ways.

## MANUAL CLEANING IN EXCEL

### TIME CONSUMING

- ✗ Remove duplicates manually cell by cell ✗
- ✗ Fix date formats manually every month ✗
- ✗ Split text columns using Text-to-Columns wizard ✗
- ✗ Delete blank rows line by line ✗
- ✗ Change data types manually cell range ✗
- ✗ Re-do all work from scratch when new file arrives ✗

## POWER QUERY AUTOMATION

### 1-CLICK REFRESH

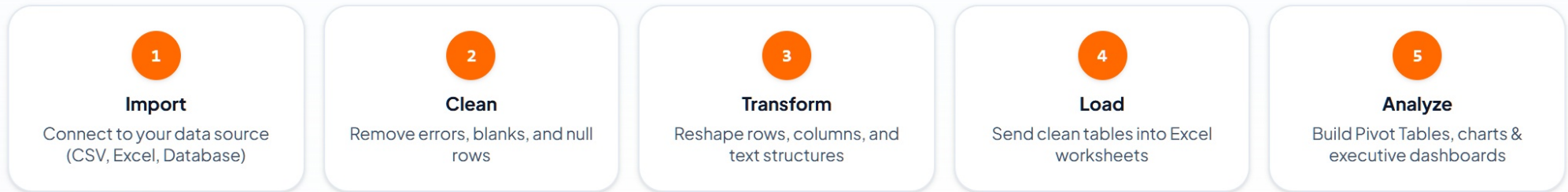
- ✓ Remove Duplicates in 1 Click ✓
- ✓ Fix Dates automatically with Data Types ✓
- ✓ Split Names dynamically by delimiter ✓
- ✓ Remove Blank Rows instantly across dataset ✓
- ✓ Enforce strict Data Types across columns ✓
- ✓ Click Refresh to re-run pipeline on new data! ✓

*"Automate repetitive data prep work and focus on strategic analysis."*

— 5 STAGES

# Power Query Workflow

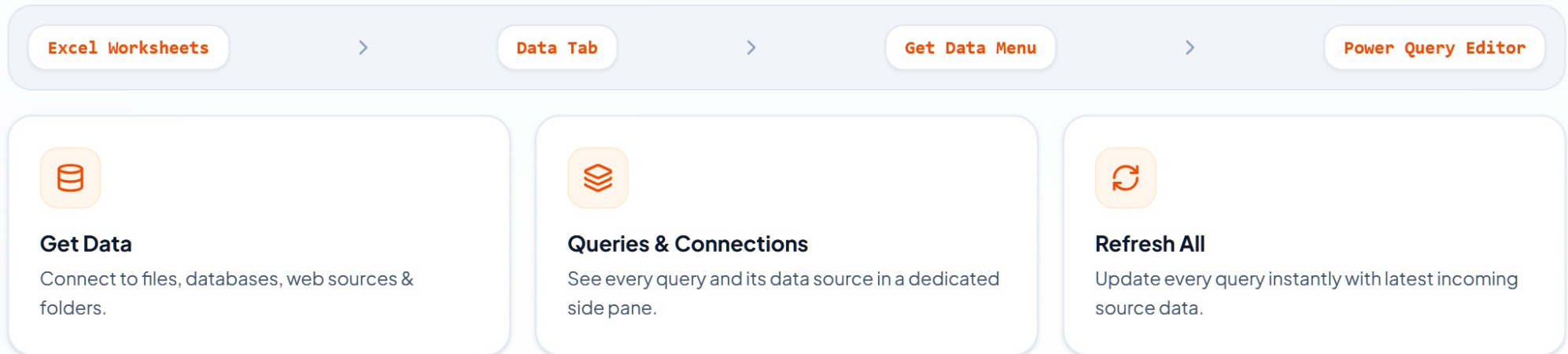
Five distinct stages from raw files to finished interactive dashboard analysis.



— NAVIGATING EXCEL

# Where Can We Find Power Query?

Access Power Query directly inside standard Microsoft Excel without installing add-ins.



— INTERFACE TOUR

# Power Query Editor Overview

One dedicated editor window, six essential functional areas to know.

## Ribbon

Commands & transformation tools organized by tab.

## Queries Pane

Lists every active query connection in workbook.

## Data Preview Grid

Live interactive preview of current cleaned table.

## Applied Steps

Sequential recorded history of every cleaning action.

## Formula Bar

Displays the underlying Power Query M code for each step.

## Column Headers

Data type indicators and quick column filter drop-downs.

— EVERYDAY TASKS

# Common Cleaning Tasks in Power Query

Everyday cleaning actions, all just a few clicks away.



## Remove Duplicates

Keep only unique data rows



## Split Columns

Break one field into several columns



## Trim Spaces

Clear extra leading & trailing spaces



## Replace Values

Swap errors or bad text for right value



## Change Data Types

Set text, number, date & boolean types



## Filter Rows

Keep only specific data rows needed



## Sort Data

Order rows ascending or descending



## Rename Columns

Give headers clear, consistent names



## Refresh Data

Re-run every step on new raw data

CORE DIFFERENCE

# Excel Formulas vs Power Query

Two different jobs — both essential for data analysts.

| Aspect       | EXCEL FORMULAS                | POWER QUERY                        |
|--------------|-------------------------------|------------------------------------|
| Scope        | Cell-based calculations       | Table-based transformations        |
| Execution    | Manual or row-by-row formulas | Automated ETL recorded steps       |
| Data Refresh | Recalculates on cell edit     | Refreshable pipeline on raw source |
| Primary Role | Calculations & Business Logic | Data Preparation & Cleaning        |

"Excel formulas calculate. Power Query prepares data."

— PRACTICAL DEMO

# Live Demo Step-by-Step Preview

What we will build together in the interactive lab, step by step.

1

## Import File

Load messy raw CSV  
customer export

2

## Remove Blank Rows

Strip empty rows across  
dataset

3

## Fix Data Types

Convert text dates and  
prices to numeric

4

## Split Full Name

Separate names into First &  
Last Name

5

## Remove Duplicates

Deduplicate records by  
Customer ID

6

## Close & Load

Deliver clean table to Excel  
worksheet

SIMULATOR LAB

# Interactive Power Query Transformation Simulator

Watch a messy raw customer table transform step-by-step through Power Query Applied Steps.

APPLIED STEPS PANE

1. Source (Raw Data)

2. Filtered Rows

3. Trimmed Text

4. Removed Duplicates

ACTION PERFORMED:

Import Raw CSV File

LIVE DATA PREVIEW GRID

3 CLEAN ROWS

| Customer ID | Full Name    | Join Date  | Status |
|-------------|--------------|------------|--------|
| 101         | john smith   | 2024/01/15 | null   |
| 102         | MARY JOHNSON | 2024-01-16 | null   |
| 101         | john smith   | 2024/01/15 | null   |

Previous Step

Next Step

— DATA TYPES

# Data Types & Error Handling in Power Query

Enforce strict column data types to guarantee calculation accuracy.

## Data Types & Error Handling in Power Query

Enforce strict column data types to guarantee calculation accuracy.

BEST PRACTICES

# Best Practices for Formulas & Power Query

Essential rules to follow for error-free, automated Excel reporting models.

✓ Always clean and prepare data in Power Query BEFORE applying complex formulas.

✓ Always specify exact match (FALSE or 0) when using VLOOKUP or HLOOKUP.

✓ Prefer XLOOKUP over VLOOKUP whenever Excel 365 / 2021 is available.

✓ Wrap risky lookup formulas with IFERROR or IFNA to prevent #N/A display.

✓ Document complex nested formulas with column header notes for team clarity.

— SUMMARY

# Key Takeaways from Session 05

Core principles to retain as you progress toward advanced analytics mastery:



01

## Saves Hours of Manual Work

Text functions & Power Query eliminate tedious line-by-line cell cleaning.



02

## Automates Repetitive Workflows

Click Refresh to run recorded ETL transformations on new monthly data files.



03

## Improves Data Quality

Enforces strict data types, removes whitespace, and strips duplicates automatically.



04

## Flexible Lookup Architecture

Mastering XLOOKUP and INDEX/MATCH enables lookups in any direction across tables.



05

## Dynamic Array Power

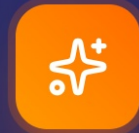
FILTER, UNIQUE, and SORT create live, auto-expanding reports without VBA.



06

## Essential Data Analyst Skill

Power Query is the cornerstone of modern Excel, Power BI, and SQL analytics pipelines.



# From Formula to Insight.

Every function and query transformation you learned today turns a spreadsheet column into a decision someone can act on.

**Next session: Power Query Deep Dive (Merge, Append, Unpivot & Group By)**

Instant Academy • Data Analysis Training Program